Diagonal Traverse (medium)

Problem Statement

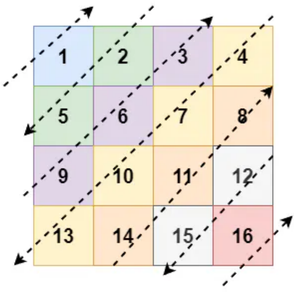
Given a 2D matrix of size `m x n`, return a 1D array containing elements of the matrix in the diagonal order.

Example 1:

- **Input**: `matrix =

```
[[1,2,3],
 [4,5,6]]
```

- **Expected Output**: `[1,2,4,5,3,6]`
- **Justification**: Traversal begins at 1, moves to 2, diagonally down to 4, up to 5, down to 3, and finally to 6.



Example 2:

- **Input**: matrix =

```
[[1,5],
 [3,6]]
```

- **Expected Output**: `[1,5,3,6]`
- **Justification**: The traversal starts from 1, moves to 2, then down to 3, and finally to 4.

Example 3:

- **Input**: `matrix =

```
[[1],
 [2],
 [3]]
```

- **Expected Output**: `[1,2,3]`
- **Justification**: The traversal is straightforward down the single-column matrix.

Try it yourself

Try solving this question here:

Java

```java
import java.util.ArrayList;
import java.util.List;

public class Solution {
    public List<Integer> findDiagonalOrder(int[][] matrix) {
        List<Integer> result = new ArrayList<>();
        // ToDo: Write Your Code Here.
        return result;
    }
}
```

Test Case | Run Code Result | Stdout

[[1,2],[3,4]]

Use Example Testcases | Run | Submit